

SPATIAL SHOCK-GATED MIXTURE OF EXPERTS FOR TRANSONIC SURFACE-LOAD PREDICTION

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Summary. AeroSurrogate is a physics-aware machine learning surrogate for the surface coefficients $[C_p, C_{fx}, C_{fy}, C_{fz}]$ on a transonic transport configuration, requiring no CFD solver at inference. An eight-layer message-passing graph neural network over a k -nearest-neighbour graph of the surface maps every node to a latent read by a shock indicator, a shock-gated Mixture of Experts for C_p and a friction head. The shock expert adds a residual scaled by the predicted shock probability, an explicit mechanism for the discontinuity: ablating it triples the C_p mean absolute error on shock nodes, from 0.0458 to 0.1370. Shock labels derive from real RANS pressures via the critical pressure coefficient, not from isentropic pseudo-labels. Trained on the ONERA CRM WBPN database and fine-tuned head-only with the backbone frozen, it reaches $R^2 = 0.9729$ for C_p and 0.9027–0.9209 for the skin-friction components over 36 508 360 held-out surface points from 140 simulations excluded from model selection.

1 INTRODUCTION

In transonic flight, normal shocks on wing surfaces produce strong adverse pressure gradients that drive boundary-layer thickening, separation and buffet onset, and so govern the drag-divergence Mach number and the structural loads envelope [1, 2]. Reynolds-Averaged Navier–Stokes (RANS) solvers resolve that structure at $\mathcal{O}(10^3\text{--}10^4)$ CPU-hours per operating condition, which rules them out of high-frequency pipelines such as shape optimisation or real-time loads estimation. Machine learning surrogates are the practical

alternative [4, 5], but a single smooth network trained across subsonic, transonic and separated regimes learns an interpolant that blurs the shock and underestimates peak suction, because the smeared solution lowers the average error over the predominantly smooth remainder of the surface.

Existing machine learning shock indicators work at the volumetric field level, from neural troubled-cell indicators [6] to their convolutional successors [7]. They require volumetric data, inaccessible in the surface-only setting of experimental aerodynamics or flight test, and commonly rely on pseudo-labels from isentropic assumptions on the inputs, which renders the problem circular. Physics-informed neural networks [8] avoid the pseudo-labels but need the governing equations embedded explicitly.

This work addresses both gaps with AeroSurrogate, a 4.6×10^6 -parameter spatial architecture predicting $[C_p, C_{fx}, C_{fy}, C_{fz}]$ at every mesh node from surface geometry and flight condition alone: shock detection becomes a neighbourhood operation on a graph-network latent, the discontinuity receives a dedicated expert, and the labels come from real RANS surface pressures rather than from an isentropic assumption.

2 DATASET AND PROBLEM FORMULATION

2.1 Configuration and simulation database

The ONERA Common Research Model Wing–Body–Pylon–Nacelle (CRM WBPN) [1, 2] is an industry-standard benchmark for transonic transport aerodynamics. This work builds on the CRM WBPN machine-learning database [3]: 468 steady RANS simulations spanning $M \in [0.30, 0.96]$, $\alpha \in [-15^\circ, +15^\circ]$ and stagnation pressures $\Pi \in \{1, 2, 4\} \times 10^5$ Pa, each supplying the target $\mathbf{y} = [C_p, C_{fx}, C_{fy}, C_{fz}] \in \mathbb{R}^4$ at every wall point. Following the ONERA regression challenge the split is at the simulation level, 312 training conditions against 156 held out; the sixteen simulations on which early stopping selects the checkpoint are drawn from the test partition, so every metric reported here uses the remaining 140 (Section 3.4). Features and targets are standardised (z-score) with training-set statistics, the edge features of Section 3.1 excepted.

2.2 Feature engineering

Each surface point carries the sixteen features of Table 1: nine base variables, five compressibility features computed analytically from them, and two geometry features, supplied explicitly because the shock on a transonic transport wing sits at a characteristic chord fraction and strengthens over the mid-span region. The last of these, the critical pressure coefficient $C_p^{\text{crit}}(M)$, is the isentropic value of C_p at which the local flow first reaches $M_{\text{local}} = 1$; it is the threshold used for the shock labelling below. No solver output enters at this stage.

Table 1: Per-node feature set $\mathbf{x} \in \mathbb{R}^{16}$; the physics and geometry blocks are computed from the base block alone, with $\gamma = 1.4$ for air.

Block	Features
Base (9)	x, y, z , coordinates normalised by the reference chord; n_x, n_y, n_z , outward unit normal; Mach number M ; angle of attack α (deg); total pressure $\Pi \times 10^{-5}$
Physics (5)	$q_{\text{dyn}} = M^2/2$; $\Pi_{\text{norm}} = \Pi/(1 + \frac{\gamma-1}{2}M^2)$; $\text{AoA}_{\text{sin}} = \sin \alpha$; $L_{\text{factor}} = \sqrt{1 - M^2}/(1 + \frac{\gamma-1}{2}M^2)$, the Laitone compressibility scale; $C_p^{\text{crit}} = (2/\gamma M^2)[(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1}M^2)^{\gamma/(\gamma-1)} - 1]$
Geometry (2)	$x_{\text{norm}} \in [0, 1]$, chord-wise position, 0 at the leading edge; $\eta \in [0, 1]$, span-wise station, 0 at the root

2.3 Shock label generation from RANS output

A binary shock label is defined for each surface point from the RANS solver output:

$$y_{\text{shock}} = \mathbf{1}[C_p^{\text{RANS}} < C_p^{\text{crit}}(M)] \wedge \mathbf{1}[M > 0.75] \quad (1)$$

The first factor marks the footprint of the supersonic pocket terminated by the shock, the second restricts the label to the transonic envelope. Because C_p^{RANS} is a solver output rather than an isentropic estimate from the inputs, the labelling is not circular; the labels are used only in training. The same $M > 0.75$ threshold separates the transonic and subsonic subsets of Section 4, whose shock subset is the transonic points that also satisfy $C_p^{\text{RANS}} < C_p^{\text{crit}}(M)$, 18.7% of the held-out partition (6 830 670 of 36 508 360).

3 METHODOLOGY: THE AEROSURROGATE ARCHITECTURE

AeroSurrogate, shown in Figure 1, is a single end-to-end trainable model: an eight-layer message-passing GNN over a per-simulation k -nearest-neighbour graph maps each node to a 256-dimensional latent $\mathbf{h}_i$ feeding a shock indicator, a shock-gated mixture of experts for C_p and a friction head; 3 472 640 of its 4 603 918 parameters lie in the backbone.

3.1 Spatial backbone

The graph is built once per simulation: its $N = 260\,774$ nodes are the surface mesh points carrying the sixteen features of Section 2, its edges the $k = 8$ nearest neighbours of the node coordinates, $E = 8N = 2\,086\,192$ directed edges from a KD-tree, so no mesh connectivity is required and a bare point cloud suffices. Each edge carries $\mathbf{e}_{ij} = (\delta x, \delta y, \delta z, \|\delta\|, \mathbf{n}_i \cdot \mathbf{n}_j)$. A node encoder $16 \rightarrow 128 \rightarrow 256$ lifts $\mathbf{x}_i$ to $\mathbf{h}_i \in \mathbb{R}^{256}$ and an edge encoder $5 \rightarrow 64 \rightarrow 128$ lifts $\mathbf{e}_{ij}$; each of the eight message-passing layers then applies a residual update normalised by LayerNorm [12], messages being aggregated at the destination by concatenated mean and maximum:

$$\mathbf{m}_{i \rightarrow j} = \phi_m([\mathbf{h}_i \parallel \mathbf{e}_{ij}]), \quad \mathbf{h}_j \leftarrow \text{LN}\left(\mathbf{h}_j + \phi_u([\mathbf{h}_j \parallel \text{mean } \mathbf{m} \parallel \max \mathbf{m}])\right) \quad (2)$$

with $\phi_m : 384 \rightarrow 256 \rightarrow 256$ and $\phi_u : 768 \rightarrow 256 \rightarrow 256$. Eight layers give an eight-hop receptive field, so each latent summarises a patch of surface rather than a point. All

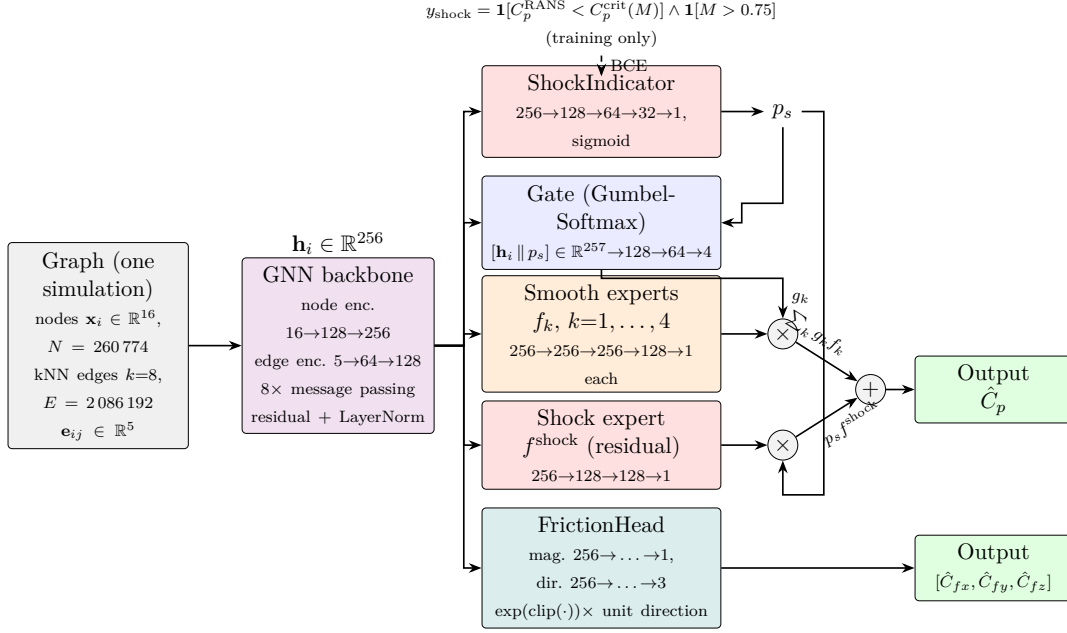


Figure 1: AeroSurrogate architecture and data flow (4603918 parameters): GNN backbone, ShockIndicator, shock-gated C_p mixture of experts, and friction head. Dashed elements exist only at training time (BCE supervision).

hidden layers use LayerNorm, SiLU [13] and dropout 0.1, the gate excepted.

3.2 ShockIndicator and shock-gated experts

The ShockIndicator is an MLP $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$ with a sigmoid output $p_{s,i} \in [0, 1]$, supervised by y_{shock} , Eq. (1), through the cross-entropy term of Eq. (5). Reading the backbone latent rather than the raw point features makes it spatial: it conditions on the pressure-gradient structure that defines a shock, and since the label is a conjunction it detects the transonic shock regime rather than local supersonic flow. On eight held-out test simulations it attains precision 0.8964, recall 0.9927 and $F_1 = 0.9421$.

The ShockGatedMoE predicts the pressure coefficient alone. Its gate maps $[\mathbf{h}_i \parallel p_{s,i}] \in \mathbb{R}^{257}$ through $257 \rightarrow 128 \rightarrow 64 \rightarrow 4$ to $K = 4$ routing logits $\mathbf{l}_i$; each smooth expert is an MLP $256 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 1$ and the shock expert a smaller $256 \rightarrow 128 \rightarrow 128 \rightarrow 1$ entering as an additive residual scaled by p_s :

$$\hat{C}_{p,i} = \sum_{k=1}^K g_k(\mathbf{h}_i, p_{s,i}) f_k(\mathbf{h}_i) + p_{s,i} f^{\text{shock}}(\mathbf{h}_i) \quad (3)$$

The residual splits the loading into a smooth background and a shock perturbation whose sharpness is inherited from the sigmoid of the indicator: it vanishes where $p_s \approx 0$ and supplies the learned pressure jump inside the supersonic pocket, as in sensor-activated

shock capturing [7], except that here both sensor and correction are learned. Because the gate is per node its specialisation is spatial rather than per condition: the mean weight of expert E_0 rises from 0.1451 off shock to 0.4722 on labelled shock nodes, emerging from routing rather than from explicit labels [9, 10]. In training the gate uses the Gumbel-Softmax relaxation [11], $\mathbf{g}_i = \text{softmax}((\mathbf{1}_i + \mathbf{G}_i)/\tau)$ with $G_k = -\log(-\log u_k)$ and τ annealed from 1.0 to 0.3 over 200 epochs; at inference it is a deterministic softmax, close to one-hot at a mean per-node entropy of 0.080 against $\log K = 1.3863$.

3.3 Friction head

The friction vector is predicted by a head reading $\mathbf{h}_i$ in parallel with the C_p branch and factorised into a magnitude and a unit direction, since $\|\mathbf{C}_f\|$ spans orders of magnitude over the surface: branches $256 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 1$ and $256 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 3$ give a log-magnitude, clipped and exponentiated, and an L_2 -normalised direction,

$$\hat{\mathbf{C}}_{f,i} = \exp(\text{clip}(f_{\text{mag}}(\mathbf{h}_i), -10, 5)) \frac{f_{\text{dir}}(\mathbf{h}_i)}{\|f_{\text{dir}}(\mathbf{h}_i)\|} \quad (4)$$

The head is not gated by p_s , so the discontinuity mechanism acts on the pressure alone.

3.4 Loss function and training

Training minimises a four-term objective over the N nodes of one simulation graph:

$$\begin{aligned} \mathcal{L} = & w_f \underbrace{\frac{1}{N} \sum_{i=1}^N w_i (\hat{C}_{p,i} - C_{p,i})^2}_{\text{shock-weighted } C_p \text{ MSE}} + \lambda_s \underbrace{\text{BCE}(\text{shock_logit}, y_{\text{shock}})}_{\text{indicator supervision}} \\ & - \lambda_{\text{lb}} \underbrace{H(\bar{\mathbf{g}})}_{\text{load balance}} + \lambda_{\text{fric}} w_f \underbrace{\mathcal{L}_{\text{fric}}}_{\text{friction}} \end{aligned} \quad (5)$$

with load balance $H(\bar{\mathbf{g}}) = -\sum_k \bar{g}_k \log \bar{g}_k$, $\bar{g}_k = N^{-1} \sum_i g_{i,k}$, and friction term

$$\mathcal{L}_{\text{fric}} = \frac{1}{N} \sum_{i=1}^N (\log \|\hat{\mathbf{C}}_{f,i}\| - \log \|\mathbf{C}_{f,i}\|)^2 + \frac{1}{N} \sum_{i=1}^N \min\left(\frac{\|\mathbf{C}_{f,i}\|}{\|\mathbf{C}_f\|}, 5\right) (1 - \cos(\hat{\mathbf{C}}_{f,i}, \mathbf{C}_{f,i})) \quad (6)$$

Here $w_i = 1 + \lambda_{\text{sw}} y_{\text{shock},i}$, $\lambda_s = 0.1$, $\lambda_{\text{lb}} = 0.01$, $\lambda_{\text{sw}} = 5.0$ and $\lambda_{\text{fric}} = 3.0$, so shock nodes carry $6\times$ weight in the C_p term and none in the friction one; $w_f \in \{1.0, 0.5\}$ is the ONERA regression-challenge confidence weight shipped with the database, which scales the two regression terms alone and makes the weighted R^2 of Section 4 comparable with the reference results. The load-balance entropy is maximised, that is, subtracted from the minimised objective: it vanishes when the gate collapses onto a single expert, so adding it instead drives the very collapse it is meant to prevent (Section 4); here it settles at 1.3827, 99.7% of uniform.

Training runs in two stages, stage A end-to-end from scratch and stage B a head-only fine-tune resuming from that checkpoint with the backbone frozen, which is the reported model. Optimisation uses AdamW, peak learning rate 10^{-3} after three warmup epochs cosine-annealed to 10^{-6} , weight decay 10^{-5} , gradient-norm clipping at 1.0, and up to 200 epochs per stage with patience 8; activation checkpointing makes one simulation graph a single optimiser step. Checkpoints are selected on the unweighted $R^2(C_p)$ over the 16 validation simulations drawn from the test split; the point-wise baseline is quoted from its own protocol, a 10% random sample of all 156 test simulations, 4 068 074 points, so its column of Table 2 is not measured on the same population.

At inference no solver output is required: the model consumes the surface nodes and the flight condition, from which every feature follows analytically, and returns the four coefficients in one forward pass with $p_{s,i}$ as an interpretable by-product. It consumes the surface as a whole, however, and so cannot score an isolated point, a deployment regression relative to the point-wise formulation.

4 RESULTS AND DISCUSSION

4.1 Aerodynamic coefficient prediction

Table 2 reports the coefficient-of-determination scores over the 140 held-out simulations and 36 508 360 surface points. In physical units the mean absolute error on C_p is 0.0276 globally and 0.0405 on the shock subset, and the confidence-weighted R^2 in which form the reference results for this database are reported is 0.9768 for C_p . In the table, nearest cond. is the no-learning baseline described below, collapsed gate is the run trained before the load-balancing sign correction, from scratch is joint end-to-end training with the corrected loss, and two-stage is the reported model: end-to-end pre-training then a head-only fine-tune with the backbone frozen.

The nearest-training-condition baseline performs no learning at all: for each held-out simulation it copies the surface field of the training condition closest to it in the normalised (M_∞, α, Π) space. The median normalised distance to that neighbour is 0.0333 and the maximum 0.1000, so the sampling is dense enough for the baseline to be demanding rather than a straw man; it reaches $R^2(C_p) = 0.9159$ globally, which places most of the aggregate score within reach of interpolation between neighbouring conditions. Where it fails is where the architecture is aimed: 0.7779 on the shock subset against 0.9350, and 0.8078 on the subsonic subset against 0.9471. A neighbouring condition puts the shock almost in the right position, and is penalised on both sides of the offset.

The gap between pressure and friction accuracy is structural: surface pressure is set by the outer, essentially inviscid flow and is a largely local function of the input vector, whereas skin friction is a wall-gradient quantity, two to three orders smaller, encoding the upstream history of the boundary layer, non-local information that eight rounds of message passing over an isotropic $k=8$ graph propagate only in part. Every regressor evaluated on this database shows the same ordering [3].

Table 2: Coefficient of determination on the held-out test set: 140 simulations (36 508 360 surface points) disjoint from both training and model selection. Columns are defined in the text; the two-stage column is the reported model and the point-wise one is measured on a different protocol.

Subset	Coef.	nearest cond.	point-wise	collapsed gate	from scratch	two-stage
global	$R^2(C_p)$	0.9159	0.9506	0.9549	0.9580	0.9729
	$R^2(C_{fx})$	0.8729	0.8512	0.8665	0.8775	0.9172
	$R^2(C_{fy})$	0.8823	0.8463	0.8443	0.8509	0.9027
	$R^2(C_{fz})$	0.8629	0.8555	0.8796	0.8806	0.9209
shock	$R^2(C_p)$	0.7779	0.8827	0.8814	0.8956	0.9350
	$R^2(C_{fx})$	0.8524	0.8390	0.8231	0.8445	0.8923
	$R^2(C_{fy})$	0.8843	0.8292	0.7867	0.8063	0.8692
	$R^2(C_{fz})$	0.9047	0.8643	0.8592	0.8677	0.9076
transonic	$R^2(C_p)$	0.9600	0.9629	0.9669	0.9698	0.9842
subsonic	$R^2(C_p)$	0.8078	0.9236	0.9259	0.9303	0.9471

The three trained columns of Table 2 separate two independent corrections, and they are not the same size. The load-balance term was an entropy that had been added to, rather than subtracted from, the minimised objective, and it therefore drove the gate toward the collapse it was meant to prevent; repairing its sign, with the training schedule held fixed, moves global $R^2(C_p)$ from 0.9549 to 0.9580 and the shock-subset score from 0.8814 to 0.8956. Adding the two-stage schedule on top of the corrected loss is worth substantially more, 0.9580 to 0.9729 globally and 0.8956 to 0.9350 on shock nodes, about $2.8\times$ the sign correction inside the shock region. The collapsed-gate run reaches only 0.8814 on shock C_p , no better than the 0.8827 of the point-wise predecessor on its own sample: the mixture pays for itself only once the gate is balanced.

Suppressing the additive shock residual $p_s f^{\text{shock}}(\mathbf{h}_i)$ at inference, with every weight otherwise unchanged, raises the C_p mean absolute error on the shock nodes of eight held-out test simulations from 0.0458 to 0.1370: the residual removes 66.6% of the error the smooth mixture alone would incur there, and on those same nodes it carries 43.4% of the magnitude of the smooth output. That is the discontinuity mechanism measured directly rather than inferred from the aggregate scores. The schedule comparison, by contrast, rests on a single run per arm, matched in data, loss and setup and both stopped on patience, so it cannot separate a curriculum effect from a fortunate initialisation.

The model scores better in the transonic regime than in the subsonic one ($R^2(C_p) = 0.9842$ against 0.9471), which follows from the objective: the shock label requires $M_\infty > 0.75$, so the six-fold weight w_i can never reach a subsonic node, and just over a quarter of the held-out transonic points carry the label (6 830 670 of 26 077 400).

4.2 How far the shock criterion compresses

Because the indicator is supervised on a physical label, it is fair to ask whether it can be written down. For the point-wise predecessor of this model, whose indicator read the nine interpretable node variables directly, symbolic regression recovers the five-node expression $p_s = \tanh(\exp(C_p^{\text{crit}}(M)/\eta))$, η the normalised span station, which activates with Mach and towards the tip. Substituting it there costs 0.0085 in $R^2(C_p)$ but 0.1005 in $R^2(C_{fy})$: the pressure criterion survives compression, the lateral friction does not.

Against the spatial indicator the same search does not close. It recovers the same $\exp(C_p^{\text{crit}}/\sqrt{\eta})$ core, which corroborates the earlier expression independently, yet a 29-node expression reaches only $R^2 = 0.4095$ against the indicator’s own output where an unconstrained regressor on the identical variables reaches 0.8850. The information is local; the closed form is what cannot hold it. Once the indicator sees an eight-hop neighbourhood its criterion stops being compactly expressible, which is the same richness that lifts the shock-subset score.

4.3 Full-aircraft surface fields

Figures 2 and 3 compare the RANS reference, the AeroSurrogate prediction and the signed error (CFD – model) for C_p and for the skin-friction magnitude $|\mathbf{C}_f|$, over both aircraft surfaces, for six held-out flight conditions spanning the envelope from $M=0.30$, $\alpha = -6^\circ$ to $M=0.90$, $\alpha=6^\circ$, none of which took any part in model selection. All six rows share a single field scale and a single error scale, so panels may be compared across conditions; Figure 4 shows the most demanding of the six on its own.

The predicted fields reproduce the characteristic structure of each regime. The two subsonic rows show the loading redistribution toward the lower surface induced by the negative incidence, without spurious oscillations; from the third row onward the upper-wing supersonic plateau appears at the correct chord-wise and span-wise station and, as Mach and incidence increase down the figure, strengthens, extends outboard and moves aft, together with the marked drop of $|\mathbf{C}_f|$ over the same region in Figure 3. The shock-labelled node fraction rises from 0.0% in the two subsonic rows to 29.3% in the bottom one, and the surrogate tracks the displacement throughout.

The error maps are signed rather than relative, since an error normalised by the local coefficient saturates wherever that coefficient crosses zero. The error is not largest where the shocks are: the two visibly worst rows are the two lowest-Mach conditions, neither of which carries a shock-labelled node. Within each row it concentrates at the wing–body junction, the inboard wing and thin bands along the leading and trailing edges, where the surface field varies over a length scale comparable to the node spacing and therefore over few graph hops; over the outer wing, including the supersonic plateau itself, it is close to zero.

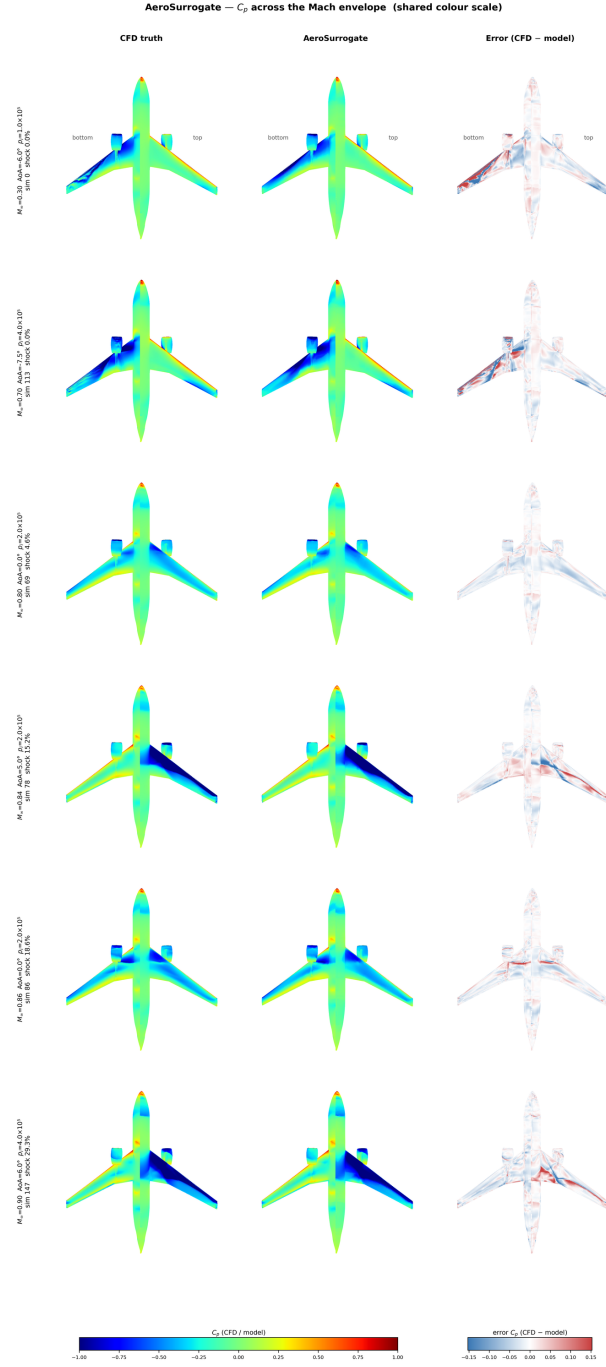


Figure 2: Pressure coefficient over six held-out flight conditions, one per row: CFD reference (left), AeroSurrogate prediction (centre) and signed error, CFD — model (right), on scales shared by all rows. Left half-model is the lower surface, right half-model the upper. Row labels give M_∞ , α , total pressure Π (printed p_t), simulation index and shock-labelled node fraction.

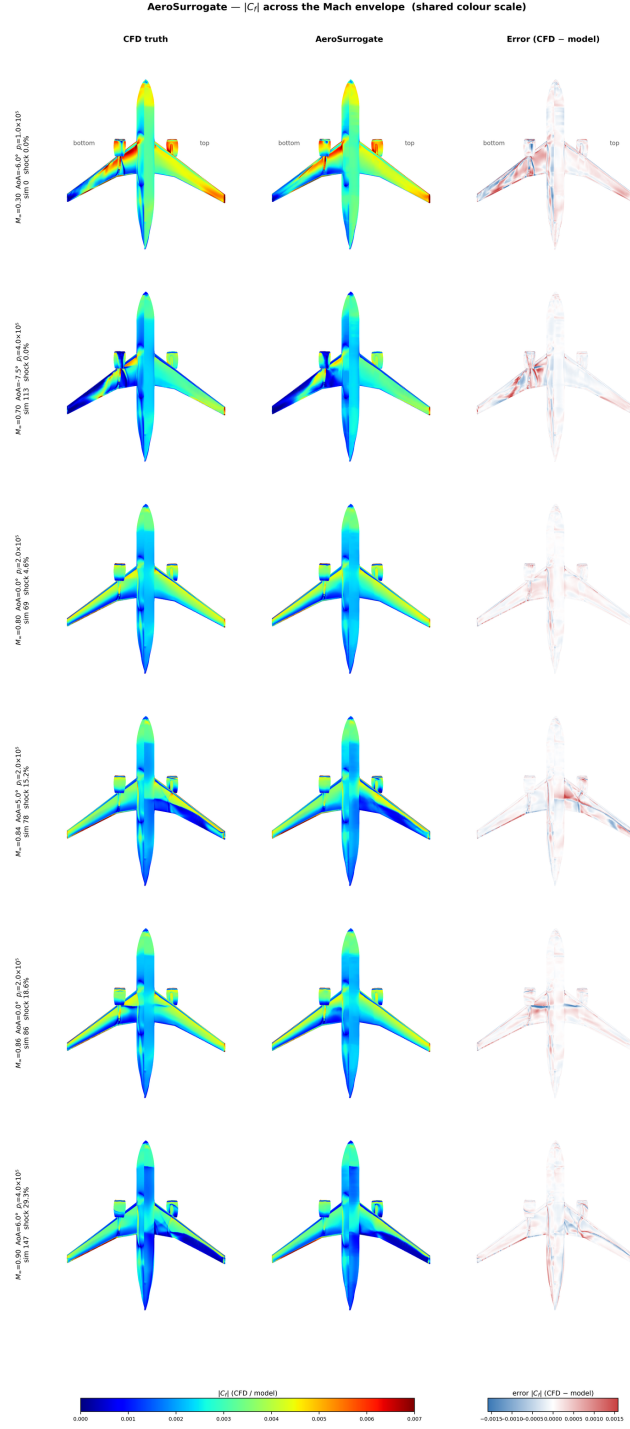


Figure 3: Skin-friction magnitude $|C_f| = \|(C_{fx}, C_{fy}, C_{fz})\|_2$ over the same six held-out conditions and in the same layout as Figure 2.

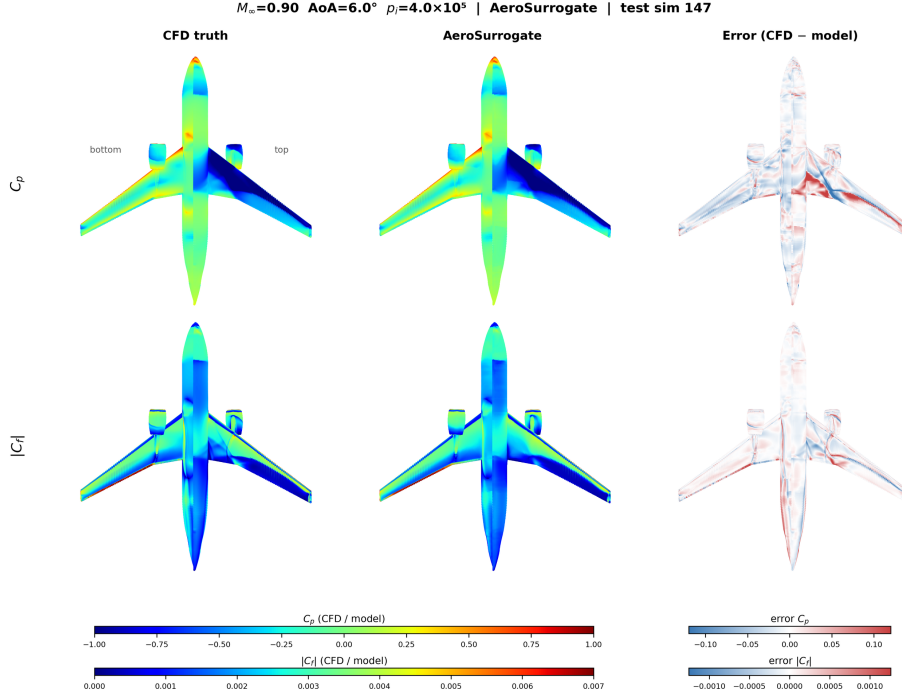


Figure 4: The most demanding condition of the six at full resolution ($M_\infty=0.90$, $\alpha=6^\circ$, $\Pi=4\times 10^5$, test simulation 147; 29.3% of its nodes are shock-labelled). Each panel places the lower surface left of the upper surface.

5 CONCLUSIONS

A physics-aware surrogate for transonic aerodynamic surfaces has been presented, coupling a spatial graph backbone over the surface mesh to a shock-gated Mixture of Experts head for C_p and a dedicated friction head. On the 140 held-out simulations, 36 508 360 surface points, it reaches $R^2(C_p) = 0.9729$ globally and 0.9350 on shock nodes. That margin is not an artefact of a densely sampled envelope: a nearest-training-condition baseline reaches 0.9159 globally but only 0.7779 on shock nodes, so the architecture earns its place where the physics is hard. Deleting the shock residual raises the shock-node C_p mean absolute error from 0.0458 to 0.1370; correcting the load-balancing sign is worth +0.0142 on the shock subset, and maturing the backbone before fine-tuning the head a further +0.0394. Labels read off the RANS pressures keep inference solver-free and are still recovered with precision 0.8964 and recall 0.9927.

Future work will repeat the two-stage and from-scratch arms over several initialisations, to separate the curriculum effect from seed variance, and revisit the closed-form criterion of Section 4.2 for the spatial indicator.

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REFERENCES

- [1] J.C. Vassberg, M.A. DeHaan, S.M. Rivers and R.A. Wahls. Development of a common research model for applied CFD validation studies. AIAA Paper 2008-6919, 2008.
- [2] D.W. Levy et al. Summary of data from the fifth CFD drag prediction workshop. *Journal of Aircraft*, Vol. 51, No. 4, pp. 1194–1213, 2014.
- [3] J. Peter, Q. Bennehard, S. Heib et al. ONERA’s CRM WBPN database for machine learning activities, related regression challenge and first results. *Computers & Fluids*, 2025.
- [4] Z.-H. Han, R. Zhang and C.-Z. Song. Aerodynamic shape optimization of natural-laminar-flow wing using surrogate-based approach. *Aerospace Science and Technology*, Vol. 74, pp. 51–60, 2018.
- [5] S. Bhatnagar, Y. Afshar, S. Pan, K. Duraisamy and S. Kaushik. Prediction of aerodynamic flow fields using convolutional neural networks. *Computational Mechanics*, Vol. 64, pp. 525–545, 2019.
- [6] D. Ray and J.S. Hesthaven. An artificial neural network as a troubled-cell indicator. *Journal of Computational Physics*, Vol. 367, pp. 166–191, 2018.
- [7] N. Discacciati, J.S. Hesthaven and D. Ray. Controlling oscillations in high-order discontinuous Galerkin schemes using artificial viscosity tuned by neural networks. *Journal of Computational Physics*, Vol. 409, 109304, 2020.
- [8] M. Raissi, P. Perdikaris and G.E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *Journal of Computational Physics*, Vol. 378, pp. 686–707, 2019.
- [9] R.A. Jacobs, M.I. Jordan, S.J. Nowlan and G.E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, Vol. 3, No. 1, pp. 79–87, 1991.
- [10] N. Shazeer, A. Mirhoseini, K. Maziarz et al. Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. *ICLR*, 2017.
- [11] E. Jang, S. Gu and B. Poole. Categorical reparameterization with Gumbel-Softmax. *ICLR*, 2017.
- [12] J.L. Ba, J.R. Kiros and G.E. Hinton. Layer normalization. *arXiv:1607.06450*, 2016.
- [13] S. Elfving, E. Uchibe and K. Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, Vol. 107, pp. 3–11, 2018.